

Algebra II with Trigonometry

Chapter 11.1

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4652951_alg-2trig-h-chp-11-1-note-docx.pdf](#)

Distance Definition A point on a parabola is exactly 7 units away from the focus. How far is this same point from the directrix?

Opening Direction Which direction (up, down, left, or right) does the parabola $y^2 = -12x$ open?

Finding the Focus What are the coordinates of the focus for the parabola $x^2 = 20y$?

Finding the Directrix If a parabola with its vertex at the origin has its focus at $(0, -6)$, what is the equation of its directrix?

Circle Equation What is the equation of a circle centered at the origin with a radius of 9?

Axis of Symmetry Is the graph of the parabola $y^2 = 16x$ symmetric about the x-axis or the y-axis?

Focal Diameter What is the focal diameter (the length of the latus rectum) of the parabola $x^2 = -32y$?

Vertex Placement If a parabola's focus is at $(4, 0)$ and its directrix is the line $x = -4$, what are the coordinates of its vertex?

Equation from Focus A parabola has its vertex at the origin and its focus at $(3, 0)$. What is its standard equation?

Latus Rectum Location The latus rectum is a line segment perpendicular to the axis of symmetry. Which specific geometric point must this line

segment pass through?